Learning to Lens

An Introduction to General Relativity
and Strong Gravitational Lensing

with Mathematica

— Instructor Edition (with Solutions) —

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Notation Guide

Symbol	Meaning	Units / Notes
Coordinates and Angles		
$\boldsymbol{\theta}$	Image position (angular)	radians or arcsec
$\boldsymbol{\beta}$	Source position (angular)	radians or arcsec
$\boldsymbol{\alpha}$	Reduced deflection angle	radians
$\hat{\alpha}$	Physical deflection angle	radians
$\boldsymbol{\xi}$	Position in lens plane	physical (kpc, Mpc)
Distances		
D_{d}	Observer–lens angular diameter distance	Mpc
D_{s}	Observer–source angular diameter distance	Mpc
D_{ds}	Lens–source angular diameter distance	Mpc
Lensing Quantities		
θ_{E}	Einstein radius	arcsec
Σ_{cr}	Critical surface mass density	$\mathrm{g\,cm^{-2}}$ or $M_\odot\,\mathrm{pc}^{-2}$
κ	Convergence $= \Sigma/\Sigma_{\mathrm{cr}}$	dimensionless
γ	Shear magnitude	dimensionless
ψ	Effective lensing potential	dimensionless
μ	Magnification	dimensionless
$\mathcal{A}$	Jacobian (inverse magnification) matrix	2×2
Cosmology		
H_0	Hubble constant	$\mathrm{km\,s^{-1}\,Mpc^{-1}}$
Ω_{m}	Matter density parameter	dimensionless
Ω_Λ	Dark energy density parameter	dimensionless
GR / Metric		
$g_{\mu\nu}$	Metric tensor	
$\Gamma^{\alpha}{}_{\mu\nu}$	Christoffel symbols	
$R^{\alpha}{}_{\beta\mu\nu}$	Riemann curvature tensor	
$R_{\mu\nu}$	Ricci tensor	
R	Ricci scalar	

Mathematica files are cross-linked throughout these notes. Links formatted as `filename.nb` or `filename.wl` point to companion Mathematica notebooks and scripts in the `Mathematica/`

directory of this repository.

Part I

General Relativity Foundations

Chapter 1

Special Relativity and Tensor Basics

> *General relativity (GR) is Einstein's theory of space, time, and gravitation. At heart it is a very simple subject . . . The essential idea is perfectly straightforward: while most forces of nature are represented by fields defined on spacetime, gravity is inherent in spacetime itself.* — Sean Carroll, *Spacetime and Geometry*

Before we can understand how gravity bends light, we need the language of special relativity and tensors. This chapter reviews the foundations: Minkowski spacetime, Lorentz transformations, four-vectors, and the index notation that will carry us through the rest of these notes. The emphasis is on building fluency with the notation and the key physical ideas — time dilation, length contraction, the invariant interval — so that the transition to curved spacetime in later modules feels natural.

Companion Mathematica files:

- `lorentz_transforms.wl` — symbolic construction and verification of Lorentz boost matrices
- `four_vectors.nb` — interactive visualizations of worldlines, light cones, and time dilation

1.1 From Newton to Einstein

In Newtonian mechanics, space and time are independent. The gravitational force between two masses M and m separated by a distance r is (cf. Carroll eq. 1.1)

$$\mathbf{F} = -\frac{GMm}{r^2}\,\hat{\mathbf{r}}\,, \tag{1.1}$$

and the gravitational potential Φ satisfies Poisson's equation,

$$\nabla^2\Phi = 4\pi G\rho\,. \tag{1.2}$$

In GR, both of these are replaced by statements about the curvature of spacetime. The key equation we will eventually arrive at is Einstein's field equation (Carroll eq. 1.5):

$$R_{\mu\nu} - \tfrac{1}{2}R\,g_{\mu\nu} = 8\pi G\,T_{\mu\nu}\,. \tag{1.3}$$

But we are getting ahead of ourselves. For now, we focus on *flat* spacetime — the arena of special relativity — and learn the tensor language needed to write down equations like (1.3).

1.2 Spacetime and the Invariant Interval

Special relativity (SR) unifies space and time into a single four-dimensional continuum called **Minkowski space**. An individual point in spacetime is an **event**, and the path of a particle through spacetime is its **worldline**.

Unlike Newtonian mechanics, SR has no absolute notion of simultaneity. Instead, the fundamental invariant is the **spacetime interval** between two events (Carroll eq. 1.10):

$$\boxed{(\Delta s)^2 = -(\Delta t)^2 + (\Delta x)^2 + (\Delta y)^2 + (\Delta z)^2} \tag{1.4}$$

where we work in **natural units** with $c = 1$ throughout this chapter (we will restore factors of c when needed for astrophysical applications).

Remark 1.2.1. We adopt the **mostly-plus** sign convention $(-, +, +, +)$, following Carroll. Some references (especially in particle physics) use the opposite convention $(+, -, -, -)$. Be careful when comparing equations across textbooks.

The interval classifies the separation between events:

- $(\Delta s)^2 < 0$: **timelike** separation (events can be causally connected)
- $(\Delta s)^2 = 0$: **null** or **lightlike** separation (connected by a light ray)
- $(\Delta s)^2 > 0$: **spacelike** separation (no causal connection possible)

The **proper time** $\Delta\tau$ along a timelike path is defined as (Carroll eq. 1.17)

$$(\Delta\tau)^2 = -(\Delta s)^2 = (\Delta t)^2 - (\Delta x)^2 - (\Delta y)^2 - (\Delta z)^2 \,. \tag{1.5}$$

This is the time measured by a clock traveling along the worldline. For an infinitesimal displacement, we write the **line element** (Carroll eq. 1.20):

$$ds^2 = \eta_{\mu\nu}\, dx^\mu\, dx^\nu \,, \tag{1.6}$$

where we have introduced the **Minkowski metric** and the **Einstein summation convention** (repeated indices are summed over).

1.3 Index Notation and the Minkowski Metric

Spacetime coordinates are labeled by Greek indices running from 0 to 3 (Carroll eq. 1.12):

$$x^\mu : \quad x^0 = t, \quad x^1 = x, \quad x^2 = y, \quad x^3 = z \,. \tag{1.7}$$

Latin indices $i, j, k, \ldots$ run over the spatial components $1, 2, 3$ only.

The **Minkowski metric** is the 4×4 matrix (Carroll eq. 1.15):

$$\eta_{\mu\nu} = \begin{pmatrix} -1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \end{pmatrix} \,. \tag{1.8}$$

Its inverse, $\eta^{\mu\nu}$, has the same components. The metric is used to **raise and lower indices**:

$$V_\mu = \eta_{\mu\nu}\, V^\nu, \qquad \omega^\mu = \eta^{\mu\nu}\, \omega_\nu \,. \tag{1.9}$$

Note that in Minkowski space this simply flips the sign of the time component: $V_0 = -V^0$, while $V_i = V^i$.

1.4 Lorentz Transformations

The coordinate transformations that leave the interval (1.6) invariant are called **Lorentz transformations**. They act on the coordinates as (Carroll eq. 1.25)

$$x^{\mu'} = \Lambda^{\mu'}{}_{\nu}\, x^{\nu}\,, \tag{1.10}$$

where the matrix Λ satisfies the defining condition (Carroll eq. 1.28):

$$\boxed{\eta = \Lambda^{\mathrm{T}}\, \eta\, \Lambda} \tag{1.11}$$

or equivalently $\eta_{\rho\sigma} = \Lambda^{\mu'}{}_{\rho}\, \Lambda^{\nu'}{}_{\sigma}\, \eta_{\mu'\nu'}$.

1.4.1 Rotations

A spatial rotation in the x–y plane by angle θ is (Carroll eq. 1.31):

$$\Lambda^{\mu'}{}_{\nu} = \begin{pmatrix} 1 & 0 & 0 & 0 \\ 0 & \cos\theta & \sin\theta & 0 \\ 0 & -\sin\theta & \cos\theta & 0 \\ 0 & 0 & 0 & 1 \end{pmatrix}. \tag{1.12}$$

1.4.2 Boosts

A **Lorentz boost** along the x-axis with rapidity parameter ϕ is (Carroll eq. 1.32):

$$\Lambda^{\mu'}{}_{\nu} = \begin{pmatrix} \cosh\phi & -\sinh\phi & 0 & 0 \\ -\sinh\phi & \cosh\phi & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \end{pmatrix}. \tag{1.13}$$

Using $v = \tanh\phi$ and $\gamma = (1 - v^2)^{-1/2} = \cosh\phi$, we recover the familiar form (Carroll eq. 1.35):

$$\begin{aligned} t' &= \gamma(t - vx)\,, \\ x' &= \gamma(x - vt)\,. \end{aligned} \tag{1.14}$$

Exercise 1.4.1. Verify that the boost matrix (1.13) satisfies the Lorentz condition (1.11) by explicit matrix multiplication. Then confirm this symbolically using `lorentz_transforms.wl`.

1.4.3 Physical Consequences

The boost equations (1.14) immediately lead to the classic relativistic effects:

- **Time dilation:** A clock at rest in S' (i.e., $\Delta x' = 0$) ticks at intervals $\Delta t = \gamma\, \Delta t'$.
- **Length contraction:** A rod at rest in S' with proper length L_0 has measured length $L = L_0/\gamma$ in S.

- **Relativity of simultaneity:** Events simultaneous in S' ($\Delta t' = 0$) satisfy $\Delta t = \gamma v \Delta x$ in S.

Exercise 1.4.2. (Twin paradox.) An observer travels from event $A = (0, 0)$ to $B' = (\Delta t/2,\, v\Delta t/2)$ and back to $C = (\Delta t, 0)$ at constant speed v, while a stationary twin travels from A directly to C. Show that the traveling twin ages by $\Delta\tau_{AB'C} = \sqrt{1 - v^2}\,\Delta t$ while the stationary twin ages by $\Delta\tau_{ABC} = \Delta t$. (cf. Carroll eqs. 1.18–1.19. Visualize with `four_vectors.nb`.)

1.5 Four-Vectors

A **four-vector** V^μ is a set of four components that transforms under Lorentz transformations as (Carroll eq. 1.39):

$$V^{\mu'} = \Lambda^{\mu'}{}_{\nu}\, V^\nu\,. \tag{1.15}$$

Key examples:

Four-position: $x^\mu = (t, x, y, z)$.

Four-velocity: For a massive particle with worldline $x^\mu(\tau)$, the four-velocity is

$$U^\mu = \frac{dx^\mu}{d\tau} = \gamma(1, v^x, v^y, v^z)\,. \tag{1.16}$$

Its norm is always $\eta_{\mu\nu}U^\mu U^\nu = -1$ (in units with $c = 1$).

Four-momentum: $p^\mu = m\,U^\mu = (E, p^x, p^y, p^z)$, where $E = \gamma m$ is the relativistic energy and $\mathbf{p} = \gamma m\,\mathbf{v}$ is the three-momentum. The invariant norm gives the **energy–momentum relation**:

$$\boxed{\eta_{\mu\nu}\, p^\mu p^\nu = -E^2 + |\mathbf{p}|^2 = -m^2} \tag{1.17}$$

or, restoring c: $E^2 = (pc)^2 + (mc^2)^2$.

For a **photon** ($m = 0$), $E = |\mathbf{p}|$ and the four-momentum is null: $p^\mu p_\mu = 0$. This will be crucial when we trace light rays through curved spacetime in later modules.

1.6 Dual Vectors (One-Forms)

Associated with every vector space is a **dual vector space** (or cotangent space T_p^*). A dual vector (or **one-form**) ω_μ transforms under the *inverse* Lorentz transformation (Carroll eq. 1.50):

$$\omega_{\mu'} = \Lambda^\nu{}_{\mu'}\,\omega_\nu\,. \tag{1.18}$$

The prototypical one-form is the **gradient** of a scalar field ϕ (Carroll eq. 1.54):

$$(\mathrm{d}\phi)_\mu = \partial_\mu\phi \equiv \frac{\partial\phi}{\partial x^\mu}\,. \tag{1.19}$$

The contraction of a vector with a dual vector is a Lorentz scalar: $\omega_\mu V^\mu \in \mathbf{R}$.

1.7 Tensors

A **tensor** of type (k, l) is a multilinear map that takes k one-forms and l vectors and produces a real number (Carroll eq. 1.56). In components, the transformation law is (Carroll eq. 1.63):

$$T^{\mu'_1 \cdots \mu'_k}{}_{\nu'_1 \cdots \nu'_l} = \Lambda^{\mu'_1}{}_{\alpha_1} \cdots \Lambda^{\mu'_k}{}_{\alpha_k} \Lambda^{\beta_1}{}_{\nu'_1} \cdots \Lambda^{\beta_l}{}_{\nu'_l} T^{\alpha_1 \cdots \alpha_k}{}_{\beta_1 \cdots \beta_l} \,. \tag{1.20}$$

1.7.1 Important Operations

- **Contraction:** Summing over one upper and one lower index reduces the rank by $(1, 1)$ (Carroll eq. 1.70): $S^{\mu\rho}{}_{\sigma} = T^{\mu\nu\rho}{}_{\sigma\nu}$.
- **Raising/lowering:** Use the metric to move indices (Carroll eq. 1.72): $T^{\alpha\beta\mu}{}_{\delta} = \eta^{\mu\gamma} T^{\alpha\beta}{}_{\gamma\delta}$.
- **Symmetrization/antisymmetrization:** Round brackets denote symmetrization; square brackets denote antisymmetrization (Carroll eqs. 1.79–1.80).

1.7.2 Key Tensors in Minkowski Space

- **Metric:** $\eta_{\mu\nu}$ — a symmetric $(0, 2)$ tensor.
- **Kronecker delta:** $\delta^{\mu}{}_{\nu}$ — the identity map.
- **Levi-Civita symbol:** $\tilde{\epsilon}_{\mu\nu\rho\sigma}$ (Carroll eq. 1.68) — totally antisymmetric, with $\tilde{\epsilon}_{0123} = +1$.
- **Electromagnetic field strength:** $F_{\mu\nu} = -F_{\nu\mu}$ (Carroll eq. 1.69) — a $(0, 2)$ antisymmetric tensor encoding $\mathbf{E}$ and $\mathbf{B}$.

1.8 The Metric as the Central Object

The key insight that connects SR to GR is that the metric is the fundamental object describing the geometry of spacetime. In SR, the metric is the constant $\eta_{\mu\nu}$. In GR, the metric becomes a field $g_{\mu\nu}(x)$ that varies from point to point, encoding the curvature of spacetime due to matter and energy.

Everything we need for gravitational lensing — the bending of light, the distortion of images, the time delays between multiple images — follows from understanding how the metric determines the paths of light rays. That story begins in the next module, where we introduce curved spacetime and the geodesic equation.

1.9 Energy and Momentum in SR

The **stress-energy tensor** $T^{\mu\nu}$ encodes the energy and momentum content of matter and fields. For a perfect fluid with energy density ρ, pressure p, and four-velocity U^{μ}:

$$T^{\mu\nu} = (\rho + p)\, U^{\mu} U^{\nu} + p\, \eta^{\mu\nu} \,. \tag{1.21}$$

Conservation of energy and momentum is expressed as

$$\partial_\mu T^{\mu\nu} = 0\,. \tag{1.22}$$

This will generalize to $\nabla_\mu T^{\mu\nu} = 0$ in curved spacetime, and the stress-energy tensor will appear as the source term on the right-hand side of Einstein's field equation (1.3).

1.10 Summary

- Spacetime is a four-dimensional continuum; the invariant quantity is the interval $ds^2 = \eta_{\mu\nu}\, dx^\mu dx^\nu$.
- Lorentz transformations ($x^{\mu'} = \Lambda^{\mu'}{}_\nu x^\nu$) leave the interval invariant. Boosts mix space and time.
- Four-vectors and tensors are classified by how they transform under Lorentz transformations.
- The metric $\eta_{\mu\nu}$ is the central object — it defines distances, raises/lowers indices, and generalizes to $g_{\mu\nu}$ in GR.
- Photons travel on null paths ($ds^2 = 0$), which will be the starting point for understanding gravitational lensing.

1.11 Problems

Exercise 1.11.1. Classify the following pairs of events as timelike, spacelike, or null separated:

(a) $A = (0, 0, 0, 0)$, $B = (2, 1, 1, 0)$

(b) $A = (0, 0, 0, 0)$, $B = (1, 1, 0, 0)$

(c) $A = (0, 0, 0, 0)$, $B = (3, 1, 2, 2)$

Check your answers with Mathematica.

Exercise 1.11.2. Show that two successive boosts along the x-axis with rapidities ϕ_1 and ϕ_2 are equivalent to a single boost with rapidity $\phi = \phi_1 + \phi_2$. Relate this to the velocity addition formula $v = (v_1 + v_2)/(1 + v_1 v_2)$.

Exercise 1.11.3. Prove that the four-velocity of any massive particle satisfies $U^\mu U_\mu = -1$ (in units with $c = 1$).

Exercise 1.11.4. A photon with energy E in frame S is observed in frame S' moving with velocity v along the x-axis.

(a) Show that the photon energy in S' is $E' = \gamma E(1 - v\cos\theta)$, where θ is the angle between the photon's direction and the x-axis in frame S.

(b) Specialize to $\theta = 0$ (head-on) and $\theta = \pi/2$ (transverse) and interpret physically.

Exercise 1.11.5. The electromagnetic field strength tensor has components (Carroll eq. 1.69):

$$F_{\mu\nu} = \begin{pmatrix} 0 & -E_1 & -E_2 & -E_3 \\ E_1 & 0 & B_3 & -B_2 \\ E_2 & -B_3 & 0 & B_1 \\ E_3 & B_2 & -B_1 & 0 \end{pmatrix}$$

(a) Show that $F_{\mu\nu}$ is antisymmetric.

(b) Compute $F^{\mu\nu} = \eta^{\mu\alpha}\eta^{\nu\beta}F_{\alpha\beta}$ and verify that the signs of the electric field components change.

(c) Compute the Lorentz scalar $F_{\mu\nu}F^{\mu\nu}$ and express it in terms of $\mathbf{E}$ and $\mathbf{B}$.

1.12 Solutions to Exercises

Full worked solutions are also available in the Mathematica script `problems_01a.wl`, which verifies each answer symbolically.

Exercise 1.11.1: Interval Classification

We compute $(\Delta s)^2 = -(\Delta t)^2 + (\Delta x)^2 + (\Delta y)^2 + (\Delta z)^2$:

(a) $(\Delta s)^2 = -(2)^2 + (1)^2 + (1)^2 + (0)^2 = -4 + 1 + 1 = -2 < 0 \Rightarrow$ **Timelike**. A massive particle can travel between these events.

(b) $(\Delta s)^2 = -(1)^2 + (1)^2 + (0)^2 + (0)^2 = 0 \Rightarrow$ **Null**. A light ray connects these events.

(c) $(\Delta s)^2 = -(3)^2 + (1)^2 + (2)^2 + (2)^2 = -9 + 1 + 4 + 4 = 0 \Rightarrow$ **Null**. Again lightlike: $|\Delta \mathbf{x}|^2 = 9 = (\Delta t)^2$.

Exercise 1.11.2: Boost Composition

The boost matrix for rapidity ϕ is $\Lambda(\phi) = \left(\begin{smallmatrix} \cosh\phi & -\sinh\phi \\ -\sinh\phi & \cosh\phi \end{smallmatrix}\right)$ (showing only the t–x block). Computing the product:

$$\begin{aligned}\Lambda(\phi_1)\,\Lambda(\phi_2) &= \begin{pmatrix} \cosh\phi_1\cosh\phi_2 + \sinh\phi_1\sinh\phi_2 & -\cosh\phi_1\sinh\phi_2 - \sinh\phi_1\cosh\phi_2 \\ -\sinh\phi_1\cosh\phi_2 - \cosh\phi_1\sinh\phi_2 & \sinh\phi_1\sinh\phi_2 + \cosh\phi_1\cosh\phi_2 \end{pmatrix} \\ &= \begin{pmatrix} \cosh(\phi_1+\phi_2) & -\sinh(\phi_1+\phi_2) \\ -\sinh(\phi_1+\phi_2) & \cosh(\phi_1+\phi_2) \end{pmatrix} = \Lambda(\phi_1+\phi_2)\,,\end{aligned}$$

using the hyperbolic addition formulas.

For the velocity addition formula, we use $v = \tanh\phi$:

$$v = \tanh(\phi_1+\phi_2) = \frac{\tanh\phi_1 + \tanh\phi_2}{1 + \tanh\phi_1\tanh\phi_2} = \frac{v_1+v_2}{1+v_1v_2}\,.$$

Note: for $v_1 = v_2 = 1$ (speed of light), $v = 2/2 = 1$, confirming that the speed of light is invariant.

Exercise 1.11.3: Four-Velocity Norm

The four-velocity is $U^\mu = \gamma(1, v^1, v^2, v^3)$ where $\gamma = (1-v^2)^{-1/2}$ and $v^2 = (v^1)^2 + (v^2)^2 + (v^3)^2$. Then:

$$\begin{aligned}U^\mu U_\mu = \eta_{\mu\nu}U^\mu U^\nu &= \gamma^2\left[-(1)^2 + (v^1)^2 + (v^2)^2 + (v^3)^2\right] \\ &= \gamma^2(-1+v^2) = \frac{-(1-v^2)}{1-v^2} = \boxed{-1}\,.\end{aligned}$$

Exercise 1.11.4: Photon Energy (Relativistic Doppler)

(a) A photon with energy E propagating at angle θ to the x-axis has four-momentum $p^\mu = E(1, \cos\theta, \sin\theta, 0)$. Applying the boost Λ along x:

$$E' = p^{0'} = \gamma(p^0 - v\,p^1) = \gamma E(1 - v\cos\theta)\,.$$

(b) Special cases:

- $\theta = 0$ (photon along $+x$, same direction as boost): $E' = \gamma E(1-v) = E\sqrt{\frac{1-v}{1+v}}$. This is a **redshift** ($E' < E$). The observer recedes from the source.
- $\theta = \pi/2$ (transverse): $E' = \gamma E = E/\sqrt{1-v^2}$. This is the **transverse Doppler effect** — a purely relativistic blueshift with no classical analogue, arising from time dilation.

Exercise 1.11.5: Electromagnetic Field Strength Tensor

(a) By inspection, $F_{\mu\nu} = -F_{\nu\mu}$ (the matrix is antisymmetric: the diagonal is zero, and off-diagonal elements differ by a sign).

(b) Raising both indices with the metric: $F^{\mu\nu} = \eta^{\mu\alpha}\eta^{\nu\beta}F_{\alpha\beta}$. In matrix form: $F^{\mu\nu} = \eta \cdot F_{\mu\nu} \cdot \eta$. The result is:

$$F^{\mu\nu} = \begin{pmatrix} 0 & E_1 & E_2 & E_3 \\ -E_1 & 0 & B_3 & -B_2 \\ -E_2 & -B_3 & 0 & B_1 \\ -E_3 & B_2 & -B_1 & 0 \end{pmatrix}$$

The electric field components flip sign (from raising the time index), while the magnetic components are unchanged.

(c) The Lorentz scalar:

$$\begin{aligned} F_{\mu\nu}F^{\mu\nu} &= 2\big(-E_1^2 - E_2^2 - E_3^2 + B_1^2 + B_2^2 + B_3^2\big) \\ &= 2(|\mathbf{B}|^2 - |\mathbf{E}|^2)\,. \end{aligned}$$

This is a Lorentz invariant: all inertial observers agree on the value of $|\mathbf{B}|^2 - |\mathbf{E}|^2$.

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